Case Study 53: Cobblestone Winery & Vineyard Estate

Campus LAN segmentation, trunking and Spanning Tree. Companion to `cobblestone_winery_case53.pkt`.

> **Status note.** The .pkt was generated by editing the saved device configs directly. It has not been run in Packet Tracer. The STP role/state table and the convergence report are predicted from the design. Confirm them with the `show` commands listed at the end and replace anything that differs.

1. VLAN, IP and port assignment

VLAN	Name	Subnet	Gateway (R1 sub-interface)	Switch ports
19	Tasting-Ops	192.168.19.0/24	.1 (Gi0/0/0.19)	SW-Tasting Fa0/4
29	Vineyard-Field	192.168.29.0/24	.1 (Gi0/0/0.29)	SW-Vineyard Fa0/3
39	Cellar-Control	192.168.39.0/24	.1 (Gi0/0/0.39)	SW-Cellar Fa0/4
49	IT-Mgmt	192.168.49.0/24	.1 (Gi0/0/0.49)	Switch SVIs: Vineyard .11, Tasting .12, Cellar .13
905	Visitor-WiFi	192.168.90.0/24	.1 (Gi0/0/0.905)	SW-Tasting Fa0/5
999	Native (unused)	none	none (native sub-interface only)	no access ports, trunk native only

Changes from the brief, with justification: VLAN 19 is defined on all three switches so the VLAN database is identical everywhere (needed for a clean VTP transparent design). VLAN 49 got management SVIs on each switch so it is actually used.

2. Trunk and EtherChannel configuration

All inter-switch links and the router uplink carry VLANs 19,29,39,49,905,999 with native VLAN 999. VLAN 1 is not allowed. Every trunk carries every VLAN so any single link failure still leaves a full path.

```
! Trunk template (all trunk ports)
 switchport trunk native vlan 999
 switchport trunk allowed vlan 19,29,39,49,905,999
 switchport mode trunk

! SW-Tasting and SW-Cellar: Fa0/2 + Fa0/3 bundled (LACP)
interface Port-channel1
 <trunk template>
interface range FastEthernet0/2 - 3
 <trunk template>
 channel-group 1 mode active
```

Switch	Trunk ports
SW-Tasting	Fa0/1 (to Vineyard), Po1 = Fa0/2+Fa0/3 (to Cellar), Gi0/1 (to R1)
SW-Vineyard	Fa0/1 (to Tasting), Fa0/2 (to Cellar)
SW-Cellar	Fa0/1 (to Vineyard), Po1 = Fa0/2+Fa0/3 (to Tasting)

Management SVIs (each switch, own address): `interface Vlan49` / `ip address 192.168.49.{11|12|13} 255.255.255.0`, plus `ip default-gateway 192.168.49.1`.

3. Inter-VLAN routing (router-on-a-stick, R1)

```
interface GigabitEthernet0/0/0
 no shutdown
interface GigabitEthernet0/0/0.19
 encapsulation dot1Q 19
 ip address 192.168.19.1 255.255.255.0
interface GigabitEthernet0/0/0.29
 encapsulation dot1Q 29
 ip address 192.168.29.1 255.255.255.0
interface GigabitEthernet0/0/0.39
```

```
 encapsulation dot1Q 39
 ip address 192.168.39.1 255.255.255.0
interface GigabitEthernet0/0/0.49
 encapsulation dot1Q 49
 ip address 192.168.49.1 255.255.255.0
interface GigabitEthernet0/0/0.905
 encapsulation dot1Q 905
 ip address 192.168.90.1 255.255.255.0
 ip access-group VISITOR-IN in
interface GigabitEthernet0/0/0.999
 encapsulation dot1Q 999 native

ip access-list extended VISITOR-IN
 deny ip 192.168.90.0 0.0.0.255 192.168.39.0 0.0.0.255
 deny ip 192.168.90.0 0.0.0.255 192.168.49.0 0.0.0.255
 permit ip any any
```

The ACL sits inbound on the visitor sub-interface, so visitor traffic to Cellar-Control is dropped before it is routed. The IT-Mgmt deny is an addition beyond the brief: visitors should not reach switch management. Remove that line if your instructor wants the minimum policy.

Verification. PC-Tasting, PC-Vineyard and PC-Cellar should all ping each other and their gateways. PC-Visitor should reach PC-Tasting and PC-Vineyard but **fail** to reach PC-Cellar (192.168.39.10).

4. Spanning Tree

Mode is PVST (classic 802.1D behaviour, so ports show as blocking and convergence is about 30 seconds). Priorities are set for all VLANs: `spanning-tree vlan 1,19,29,39,49,905,999 priority <value>`.

Switch	Priority	Reason
SW-Cellar	4096	Root bridge: it houses the fermentation-monitoring master.
SW-Tasting	8192	Deliberate secondary root: the router uplink is here, so it is the best fallback core.
SW-Vineyard	32768 (default)	Edge building, should never become root.

Expected port roles and states (predicted, confirm with `show spanning-tree`):

Switch	Port	Role	State
SW-Cellar	Fa0/1 (to Vineyard)	Designated	Forwarding
SW-Cellar	Po1 (to Tasting)	Designated	Forwarding
SW-Tasting	Po1 (to Cellar)	**Root**	Forwarding
SW-Tasting	Fa0/1 (to Vineyard)	Designated	Forwarding
SW-Vineyard	Fa0/2 (to Cellar)	**Root**	Forwarding
SW-Vineyard	Fa0/1 (to Tasting)	Alternate	**Blocking**

The Vineyard Fa0/1 port blocks because Vineyard already has a direct path to the root, and the Tasting side of that link has a lower root path cost (and, in a tie, the lower bridge priority), so Tasting wins the designated role on that segment. Access ports and the router link are edge/designated forwarding ports.

5. EtherChannel and STP's view

The two parallel Tasting-Cellar links are bundled into Po1 using LACP (`mode active` on both ends). Without the bundle STP sees two parallel paths between the same two switches and blocks one of them, wasting half the capacity. Once the bundle is active, STP treats Po1 as a single logical link: one port, one cost, one role. Both physical links forward traffic, and losing one member does not trigger an STP topology change as long as the other stays up.

6. Native VLAN 999: security reason

The native VLAN is the one VLAN whose frames cross an 802.1Q trunk untagged. If it is the default VLAN 1, an attacker on an access port in that VLAN can send a frame with two 802.1Q tags. The first switch strips the outer tag (because it matches the native VLAN) and forwards the frame with the inner tag onto the trunk, delivering it into a VLAN the attacker was never assigned to (VLAN hopping by double tagging). Using VLAN 999, which has no access ports, no hosts and no IP address, means no attacker can ever sit in the native VLAN, so the attack has no entry point. Setting the same native VLAN on both ends of every trunk also avoids mismatch problems.

7. Port security (Visitor-WiFi)

```
interface FastEthernet0/5
 switchport access vlan 905
 switchport mode access
 switchport port-security
 switchport port-security maximum 3
 switchport port-security violation restrict
 spanning-tree portfast
 spanning-tree bpduguard enable
```

Why restrict, not shutdown. This port feeds an access point that carries many visitor devices. In shutdown mode, one extra or spoofed MAC would err-disable the port and take down Wi-Fi for every visitor until staff intervene, which gives an attacker a trivial denial of service against the tasting room. Restrict drops frames from the offending MAC, logs the violation and increments the counter, while legitimate devices keep working. The trade-off is that restrict is noisier and offers no hard containment, so the logs must actually be monitored.

8. VTP recommendation memo

Recommendation: VTP transparent mode on all three switches.

Transparent mode means each switch keeps its own locally configured VLANs and ignores advertisements from others (it still forwards them). The estate has three switches and six VLANs that will rarely change. The main risk of server/client mode is the revision-number overwrite: a switch added or reconnected with a higher configuration revision number and a different VLAN database replaces the database everywhere. Here that could delete Cellar-Control (VLAN 39) across the estate and cut the fermentation sensors and climate controllers during a vintage, which is the one outcome this design exists to prevent.

Trade-off accepted. Every VLAN change must be made by hand on all three switches, and a forgotten switch silently blackholes that VLAN. At this scale that is a small, visible, auditable cost; a wiped VLAN database is a large, fast, estate-wide one. Server/client mode becomes the better choice if the network grows to many switches with frequent VLAN changes. In that case use VTP version 3, or at minimum a domain password, and always reset the revision number on any switch before connecting it.

9. Convergence report: Vineyard-Cellar link failure

Scenario. Shut SW-Vineyard Fa0/2 (or unplug the cable) while PC-Vineyard pings PC-Cellar continuously.

1. The link goes down on both ends (Vineyard Fa0/2, Cellar Fa0/1). Vineyard loses its root port.
2. Vineyard's Fa0/1 already holds a valid BPDU from the root via Tasting (it is the alternate port). Because this is a direct failure, Vineyard does not wait for max-age (20 s); it promotes Fa0/1 to the new root port immediately.
3. Fa0/1 moves through blocking, then listening (15 s forward delay), then learning (15 s), then forwarding. In classic PVST that is about 30 seconds.
4. The topology change is flooded; switches shorten MAC aging to the forward-delay value, so MAC entries are flushed and relearned.
5. Final state: SW-Vineyard Fa0/1 is root/forwarding, Tasting Fa0/1 stays designated, and the root is unchanged.

Port that changes state: SW-Vineyard Fa0/1, blocking to forwarding. No other port changes role.

Impact on cellar climate-control traffic. Cellar's own path to the router is the Po1 bundle to Tasting, which is untouched, so Cellar-Control hosts keep their gateway throughout. What is affected: (a) anything in VLAN 29 on Vineyard is cut off for about 30 seconds, including harvest tablets, and any Cellar-to-Vineyard flow stalls; (b) the topology change flushes MAC tables, so Cellar-Control unicast is briefly flooded and relearned, which can cause short loss or latency spikes. The risk to cellar sensors is therefore low in this scenario, but a failure of the Tasting-Cellar path would be worse.

Mitigation. Switch to Rapid PVST (`spanning-tree mode rapid-pvst`) to cut convergence to roughly 1 to 2 seconds. This case study kept classic PVST to match the brief's "blocking" wording.

10. Verification checklist (do this in Packet Tracer)

Wait 30 to 60 seconds after opening the file for STP and LACP to settle, then run:

- `show vlan brief` on each switch: VLANs 19,29,39,49,905,999 present, access ports in the right VLAN.
- `show interfaces trunk` : native 999, allowed list correct, Po1 listed.

- `show etherchannel summary` on Tasting and Cellar: Po1 should be `SU` with Fa0/2 and Fa0/3 flagged `P`.
- `show spanning-tree vlan 39`: SW-Cellar is the root. Capture roles and compare to the table above.
- `show port-security interface fa0/5` on Tasting: max 3, restrict.
- `show vtp status`: mode must read Transparent. If it does not, set it with `vtp mode transparent`.
- `show ip interface brief` on R1: all five sub-interfaces up.
- Ping tests from section 3, then the failure test from section 9.

If any part of the config did not load, paste the listings above into the relevant device CLI.